

REQUIREMENT ANALYSIS PHASE – TECHNOLOGY STACK

DATE	04/10/2026
TEAM ID	SWTID-2026-8904
PROJECT NAME	AI Study Buddy

MAXIMUM MARKS: 4 Marks

Technical Architecture

The architecture uses a web interface, Node.js/Express backend, MongoDB database and Google Gemini AI service. The application can run locally during development and can later be deployed to cloud services.

Table 1: Components & Technologies

No.	Component	Description	Technology
1	User Interface	Registration, login, study dashboard, notes, quiz and AI chat.	HTML, CSS, JavaScript / React if used
2	Application Logic-1	User registration, login and authentication.	Node.js, Express.js, JWT, bcryptjs
3	Application Logic-2	Create, view, update and search study materials.	Node.js, Express.js
4	Application Logic-3	Generate explanations, summaries and quizzes.	Google Gemini API
5	Database	Store user profiles, notes, quizzes and progress.	MongoDB, Mongoose
6	Cloud Database	Optional online database deployment.	MongoDB Atlas

Table 2: Application Characteristics

No.	Characteristic	Description	Technology
1	Open Source	Uses open-source backend and database technologies.	Node.js, Express.js, MongoDB
2	Security	Password hashing, JWT authentication and protected routes.	bcryptjs, JWT, HTTPS
3	Scalability	Modular REST API supports more users and study content.	Node.js, MongoDB
4	Availability	Students can access study tools when the application services are available.	Node.js server / Cloud hosting
5	Performance	Database indexing and efficient API calls support quick retrieval and AI requests.	MongoDB, Mongoose, Gemini API